1 Initialize document weights

computed once before decoding



Query *x*

How many eggs does this bird lay?

Retriever
selects set Z

knowledge base

Z

K singleton inputs

(x, z_1)

(x, z_2)

⋮

(x, z_K)

VLM
shared Θ
K parallel passes

Last-token embeddings

$e_\Theta(x,z_1)$

$e_\Theta(x,z_2)$

⋮

$e_\Theta(x,z_K)$

MLP
$s(\cdot)$
softmax

Document prior

$m_{1,k} = P_\Theta(z_k \mid x,Z)$

Eq. 5.7

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2 Decode token j and update the posterior

repeat until EOS

K parallel VLM branches

$(x,z_1,y_{<j})$

$(x,z_2,y_{<j})$

⋮

$(x,z_K,y_{<j})$

Per-document distributions

$p_{j,k}(\cdot) = P_\Theta(\cdot \mid y_{<j},x,z_k)$

Posterior-weighted mixture

$P_\Theta(y_j \mid y_{<j},x,Z)$

$= \Sigma_k\, m_{j,k}\, p_{j,k}(y_j)$

Eq. 5.4

Decode
greedy / sampling

y_j

Current document posterior

$m_{j,k} = P_\Theta(z_k \mid y_{<j},x,Z)$

Eq. 5.5

Recursive Bayes update for the next step

$m_{j+1,k} \leftarrow \text{Normalize}[\, p_{j,k}(y_j)\, m_{j,k}\,]$

Eq. 5.6

use updated posterior at step j + 1